Algorithmic Agnosticism: Why No Theory of Everything Can Be Known to Be Final

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Abstract

Physicists hope for a final theory of everything. We argue that no such theory could be known to be final, where knowing finality means that the observations rule out every consistent rival. We assume that every observation is a finite record and every usable method for choosing theories is computable. From learning theory and algorithmic information theory we show three things. First, no method of any kind can certify a theory as final: every finite record fits infinitely many computable theories that disagree about the unobserved. Second, no usable method is guaranteed to settle on a final theory: each record fits some possible world in which the method never settles on the right theory, or, if it uses chance, settles with arbitrarily small probability. Third, even an agent with unlimited data and computation cannot separate formulations with identical predictions or tell apart with certainty chance hypotheses that agree on which events have probability zero. Simplicity does not decide. The data fix which theories fit, but their lengths depend on the universal computer chosen to measure them. Rivals that fit equally well, are nearly as short and disagree about the unobserved always exist, and a change of computer that alters no length by more than about their gap makes one the shortest. Weighting theories by length predicts within as many bits of each model as it takes to describe that model, but choosing among survivors is a constraint we impose. Compact, unified theories remain worth seeking for their testable predictions. We call this position algorithmic agnosticism.

Keywords: underdetermination; scientific realism; Solomonoff induction; minimum description length; formal learning theory; physical Church–Turing thesis; theory of everything; agnosticism.

Contents

1 Introduction **2**

2 Framework and premises **4**
2.1 Records, models, worlds 4
2.2 Premises 6
2.3 The argument in outline 7

3 No record certifies a final theory **7**

4 No usable method finds one **10**

5 Simplicity does not single one out **13**
5.1 Description length as a constraint 13
5.2 The choice of machine 14
5.3 Near-minimal rivals 16
5.4 Other preferences, and serious rivals 17

6 What even an ideal agent cannot know **17**

7 What the search is good for **19**

8 Objections and replies **21**

9 Conclusion **24**

Statement on the use of AI **25**

1 Introduction

Much of fundamental physics is organized around the hope of a final theory: a single theory of everything from which every regularity follows, and which is not itself the approximation of something deeper (Weinberg, 1992). Some go further and hold that the world simply *is* such a structure (Tegmark, 2008), or that a unique computational object underlies all possible physics (Wolfram, 2021). We do not ask whether a final theory exists. We ask whether one could ever be *known* to be final (definition 2.6), and we argue that it could not.

The argument starts from two premises. The first is nearly definitional: every observation, measurement or other use of a system leaves a finite record written at finite precision (premise P1). The second is an assumption we assert and defend rather than prove: every system anyone can use, including every method of choosing among theories, is computable or a computable stochastic process (premise P2). It is the modest physical Church–Turing thesis (Piccinini, 2011), extended from devices to methods.

From these premises the main claim follows in three parts.

(C1) *No certification.* Any theory that fits every observation made so far is one of infinitely many that fit them equally well and disagree about what has not yet been observed. So no method of any kind, computable or not, can certify a theory as final: if a method certifies a model, deterministic or stochastic, at some record, a computable world that produces that record makes the certification wrong (corollary 3.2).

(C2) *No reliable discovery.* No usable method is guaranteed, even with any fixed probability, to settle on a final theory. Whatever has been observed so far, there is a possible world that fits all of it in which a deterministic usable method never settles on the right theory; for a randomized usable method and every $p > 0$, there is such a world in which it settles with probability less than p (theorem 4.1 and remark 4.2). A user can therefore never rule out living in such a world. This part rests on the second premise; for deterministic methods it needs only the weaker premise that every world computable at a method's own level of power is possible (premise P2* and theorem 4.5).

(C3) *Permanent underdetermination, even in principle.* Even an ideal agent with unlimited data and computing power faces two limits. (a) Formulations that make identical predictions

cannot be separated by any method. (b) Chance hypotheses that agree on which events have probability zero cannot be told apart *with certainty* by any method (theorem 6.1).

A corollary follows (C′): simplicity narrows the field but cannot decide among the survivors that matter. The data fix which models fit; which of them is shortest depends on the reference machine. Rivals that fit equally well, are within $K(x) + O(1)$ bits of the shortest, and disagree at an unobserved setting x always exist, and a change of machine that moves no description length by more than about that gap makes such a rival the unique shortest (proposition 5.7, theorem 5.5, and corollary 5.8). The universal mixture predicts within $K(\nu)$ bits of every model ν (theorem 7.1). Every choice among those survivors—by simplicity, beauty, naturalness or unification—is a constraint we impose (lemma 4.6). None of this makes the search for compact, unified theories pointless. Compact theories compress what is known and commit to predictions about what is not, which can be tested (section 7). What is misguided is the goal of a final theory known to be final.

Algorithmic agnosticism. We call this position *algorithmic agnosticism*.[1] Huxley, who coined the word, stated agnosticism as a method: do not pretend that conclusions are certain which are not demonstrated or demonstrable (Huxley, 1894, pp. 245–246). In the sense we intend, commonly glossed as neither affirming nor denying a claim that evidence cannot settle, we neither affirm nor deny that a final theory exists; we hold that none could be known to be one. The position differs from van Fraassen's constructive empiricism (van Fraassen, 1980), which is agnostic about unobservable entities: we are agnostic about which law generates the observations themselves. The qualifier *algorithmic* names the framework of the argument, in which theories are programs and usable methods are algorithms. For the contrast with Müller's *algorithmic idealism* (Müller, 2020, 2026), see section 8.

What we do not claim. We do not claim that the world is computable, discrete, continuous, random or deterministic, nor that it lacks a true description.

Precursors. That universal laws can in principle be refuted but never verified goes back to Hume and was made central by Popper (Popper, 1959). That a finite record admits arbitrarily many laws, so that a law is worth having only if it compresses its data, goes back to Leibniz and Weyl (Leibniz, 1686; Weyl, 1932) and was made precise by Chaitin (Chaitin, 2007). Putnam proved that no computable method extrapolates every computable sequence (Putnam, 1963), and Gold that none identifies every computable law in the limit (Gold, 1967); Kelly recast scientific method in these terms (Kelly, 1996). Assuming an objective, computable world, Hutter formalized theories of everything as programs selected by description length, and called such a theory "subjective" because it must include the observer's localization within the world (Hutter, 2010, Sec. 4). He also notes that "for every model we can construct an alternative model that behaves identically in the past" and predicts differently (Hutter, 2010, Sec. 4), which is our theorem 3.1, and he formalizes and examines a universal theory of everything that generates all computable universes (Hutter, 2010, Sec. 6); choosing among such theories by description length is machine-relative (theorem 5.3). Schmidhuber treats a theory of everything as a prior over computable universes and notes that assuming it formally describable "imposes a strong inductive bias" (Schmidhuber, 2000, abstract); lemma 4.6 makes the point general. Sterkenburg

[1] The name is unrelated to "agnostic learning" in computational learning theory (Kearns et al., 1994).

argues that Solomonoff prediction cannot justify a preference for simplicity: its predictors are mixtures over effective hypotheses, so the operative assumption is effectiveness, and the resulting "simplicity" depends on an arbitrary choice of universal machine (Sterkenburg, 2016). A strong case for simplicity is Vitányi's: ideal MDL is optimal when the data are typical for the true hypothesis, and algorithmic statistics separates a record's meaningful information from its noise (Vitányi and Li, 2000; Gács et al., 2001); we answer it in sections 5 and 8. Hawking argued from Gödel's theorem that a final theory may be out of reach (Hawking, 2002), and Faizal and colleagues argue from incompleteness that a theory of everything would require non-algorithmic understanding (Faizal et al., 2025). Kiefer concludes from Gödel's theorems that, unless space-time is fundamentally discrete, "we can never decide whether a given theory is the final one or not" (Kiefer, 2024, abstract). Our route to a similar conclusion runs through underdetermination by finite records and the limits of learning, not through undecidability within the theory, so it does not depend on the theory's arithmetic strength or on whether space-time is discrete. Hossenfelder argues that the appeal to beauty and naturalness has misled the search for a final theory (Hossenfelder, 2018).

Contributions. Little of the mathematics is new; the contribution is the assembly, aimed at a specific target. Specifically: (i) we show that the core of the claim, no certification (C1) and permanent underdetermination (C3), holds for every method whatsoever; (ii) we state Gold's theorem in a form in which, whatever record has been observed, some world that fits it is one the method never gets right, which answers the reply that a method need only succeed in *our* world; (iii) we make the machine-relativity of simplicity quantitative for theories fitted to finite data (theorem 5.3), rather than leaving it as a remark about arbitrary constants; (iv) we isolate the underdetermination that is permanent even for an ideal agent (theorem 6.1); and (v) we reach a conclusion like Kiefer's (above) from finite records and the limits of learning alone.

Outline. Section 2 fixes definitions and states and defends the premises. Section 3 shows that no record certifies a final theory, and section 4 that no usable method reliably finds one. Section 5 shows that simplicity does not single one out, and section 6 what even an ideal agent cannot know. Section 7 explains what the search is nonetheless good for. Section 8 answers objections, and section 9 concludes.

2 Framework and premises

2.1 Records, models, worlds

Fix a finite alphabet Σ with $|\Sigma| \geq 2$. Write Σ^* for the finite strings over Σ, Σ^∞ for the infinite sequences, and $u \sqsubseteq v$ for the prefix order on strings and sequences.

Definition 2.1 (Record)**.** An *observation* is a pair $(x, y) \in \Sigma^* \times \Sigma^*$: a setting x and a recorded outcome y. A *record* is a finite sequence $\big((x_0, y_0), \ldots, (x_{n-1}, y_{n-1})\big)$ of observations; repetitions are allowed, so repeated trials keep their order and multiplicity. Records grow by extension. We write R also for the underlying set of observations when order does not matter, and $\mathrm{dom}(R) = \{x_i\}_{i<n}$. A record is *functional* if no setting occurs with two distinct outcomes. Fixing a prefix-free pairing code, a record is encoded as a single string $r \in \Sigma^*$, and extension of records is prefix extension of their codes.

Fix an acceptable (Gödel) numbering $(\varphi_e)_{e\in\mathbb{N}}$ of the partial computable functions $\Sigma^* \rightharpoonup \Sigma^*$ (Rogers, 1967); write $\varphi_e(x)\!\downarrow$ when the computation halts.

Definition 2.2 (Model)**.** (a) A *deterministic model* is an index $e \in \mathbb{N}$. It is *consistent* with a functional record R if $\varphi_e(x)\!\downarrow = y$ for every $(x, y) \in R$; we write $\mathrm{Cons}(R)$ for the set of such e. Two models *diverge off* R if they give different outputs at some setting not in $\mathrm{dom}(R)$. (b) For a fixed sequence of settings, fix an effective enumeration $\nu_1, \nu_2, \ldots$ of the lower semicomputable semimeasures on outcome sequences (Li and Vitányi, 2019). A *stochastic model* is an index $i \geq 1$; $\nu_i(r)$ is the probability it assigns to outcomes beginning with r, and it is consistent with a record with outcome string r if $\nu_i(r) > 0$. (c) Two models are *extensionally equal* if they have the same extension: $\varphi_e = \varphi_{e'}$ for deterministic models, $\nu_i = \nu_j$ for stochastic ones.

Definition 2.3 (World)**.** A *world* is whatever produces the records. A world is *computable* if its record-producing behavior is given by a total computable function $f : \Sigma^* \to \Sigma^*$ (deterministic) or a computable measure μ (stochastic). We make no claim that the actual world is computable; computable worlds are the worlds that some usable model could describe exactly.

Definition 2.4 (Final model; theory of everything)**.** A model is *final* for a world if it agrees with every record the world can produce: $\varphi_e = f$ in the deterministic case, $\nu_i = \mu$ in the stochastic case. A *theory of everything* is any model offered as final.

Definition 2.5 (Method; certification; identification)**.** A *method* is a function from (codes of) records to models; a *randomized* method also reads bits from a random source. A method *certifies* a model at a record R if on R it outputs that model together with the verdict "final". Let $x_0, x_1, \ldots$ be the length-lexicographic enumeration of the settings. The *canonical presentation* of a total f is the sequence of records $R_{f\restriction n} = \big((x_0, f(x_0)), \ldots, (x_{n-1}, f(x_{n-1}))\big)$, $n \in \mathbb{N}$. A method that has already observed a record R consistent with f is presented with f *after* R: it receives the records $R \frown R_{f\restriction n}$, $n \in \mathbb{N}$ (for empty R, the canonical presentation). It *identifies* f *in the limit* after R if its outputs on this presentation are eventually constant at an index of f (Gold, 1967); it does not have to announce when this has happened. A method that is undefined on some record of the presentation does not identify f. A randomized method identifies f *with probability* p if the set of random bit sequences on which it identifies f has probability p. A method is *guaranteed* to find a final theory after R if it identifies in the limit, after R, every total computable world consistent with R; a randomized method is guaranteed with probability p if it identifies each such world with probability at least p.

Definition 2.6 (Known to be final)**.** A theory T is *known to be final*, in the sense used here, on a record R if R rules out every rival consistent with it: T is final (definition 2.4) for every computable world that can produce R: for a deterministic theory, every total computable f consistent with R; for a stochastic theory, every computable measure μ with $\mu(r) > 0$, where r is the outcome string of R. Equivalently, a method that certifies T at R is right in every such world.

We take definition 2.6 as the standard for the hope of a final theory: finality recognized from the evidence. It counts only computable worlds as rivals, so it is weaker than a standard that counts every world, and a result that defeats it defeats every stronger one.

Definition 2.7 (Constraint; selection)**.** A *constraint* (or *bias*) is given by a cost $c_R : \mathbb{N} \to \mathbb{N} \cup \{\infty\}$ on models, possibly depending on the record. Its *selection* at a functional record R is

$$\mathrm{Sel}_c(R) = \operatorname*{arg\,min}_{e\in\mathrm{Cons}(R)} c_R(e).$$

A constraint is *imposed* if the record does not determine it: other constraints, equally consistent with the record, select differently on the same record, and nothing in the record decides between them.

"Which model is final" can be asked at three levels: of *programs* (indices), of *extensions* (the behavior φ_e), and of *chance hypotheses* (measures).

2.2 Premises

Premise P1 (Finite record)**.** Every use of any system by any human or machine yields a record in the sense of definition 2.1.

Premise P1 is close to definitional. Uses take finite time; instruments have finite resolution; outcomes are written down, stored or acted upon in finite memory. Even an instrument reading a continuous quantity reports a finite-precision value. Premise P1 is a claim about *interfaces*, not about the systems behind them: it says nothing about whether the world is discrete, continuous, computable or random. *Refuted by:* a use of some system whose outcome is stored or acted on as an infinite object, for instance an instrument whose reading is available as an exact real number.

Premise P2 (Usable systems are effective)**.** Every system that any human or machine can use to produce outputs from inputs—in particular every method by which records are turned into predictions or model choices—is effective: its input–output behavior is computable or, if it is stochastic, is described by a computable probability measure on outputs.

Premise P2 is an assumption; it does not follow from premise P1. It is Piccinini's *modest* physical Church–Turing thesis (Piccinini, 2011, 2015)—every function computable by a process an observer can *use* is Turing-computable—whose usability conditions we adopt: readable inputs and outputs, a rule specifiable independently of the process, repeatability, settability, physical constructibility and reliability (Piccinini, 2015), a list he does not intend to be exhaustive. We extend it in scope only: through the measure clause to stochastic systems, and to every usable method, including methods of choosing models. A stochastic method in the sense of premise P2 is a computable machine with access to fair coin flips, the model of Pitt (1989); every computable measure on outputs is realized by such a machine.

Refuted by: a usable hypercomputer, that is, a system whose non-computable behavior can be specified, relied on and put to work.

Defense. The three standard candidates for a usable hypercomputer fail.

(a) Continuous theories. A theory stated over the reals is used by computing its consequences to finite precision, and whatever a theory yields in that way is computable in the sense of computable analysis (Weihrauch, 2000). Some continuous theories have non-computable values—the wave equation has computable initial data whose unique (weak) solution at a later time is not computable (Pour-El and Richards, 1981)—and recurrent networks with real-valued weights decide non-recursive languages (Siegelmann and Sontag, 1994). But such values cannot be used: no finite record can read them out. In the network case the non-computability comes out only where it has been put in, through non-computable weights (Davis, 2004).

(b) Relativistic hypercomputers. In Malament–Hogarth spacetimes, for example that of a rotating (Kerr) black hole, an observer could in principle receive, in finite proper time, the

result of an infinite computation (Hogarth, 1992; Etesi and Németi, 2002). But the observer cannot rule out that a "does not halt" verdict reflects spurious signals or a failure of the device rather than an infinite computation, so the output cannot be relied on; Earman and Norton, who make this point, regard such computation as an open question rather than an impossibility (Earman and Norton, 1993).

(c) Quantum randomness. A random source can produce a non-computable sequence, but it cannot be steered to compute anything specified in advance.

Theorem 2.8 (Randomness adds no power to compute functions; de Leeuw et al., 1956, Thms. 2–3)**.** *If p is a computable real and a probabilistic machine with a p-biased coin enumerates a set with positive probability, the set is computably enumerable; if it outputs $f(n)$ with probability greater than $\frac{1}{2}$ on every input n, then f is computable.*

Proposition 2.9 (Born statistics are computable)**.** *Let a finite-dimensional quantum system have a computable initial state ψ, a computable unitary U and a measurement with computable projectors $\{P_k\}$. Then the outcome probabilities $\|P_k U\psi\|^2$ (Born, 1926) are computable reals, and the outcomes of repeated independent measurements form a computable measure.*

Proof sketch. Matrix products, exponentials of computable matrices and norms are computable operations on computable reals (Weihrauch, 2000; Pour-El and Richards, 1989); a product of computable probabilities defines a computable product measure. □

A quantum random number generator is therefore a computable stochastic process in the sense of premise P2: what can be relied on about it is a computable measure, and by theorem 2.8 its randomness computes nothing new. A coin with a non-computable bias would act as an oracle, but no finite record could show that its bias is non-computable, so it could not be relied on as one.

2.3 The argument in outline

Table 1 is the map of the argument. It lists every step in the order of the sections that prove it, with the level at which each holds.
C2 uses premise P2, and for deterministic methods only its weaker form P2*. C′ uses premise P2 only to read the benchmark of T4 as unusable.

3 No record certifies a final theory

This section proves that no finite record singles out a final theory, and that no method of any kind can therefore certify one. The results concern records alone, and they hold whatever produced the records.

Theorem 3.1 (Infinite underdetermination)**.** *Let R be any finite functional record.*

(a) For every $x \notin \mathrm{dom}(R)$ and every $y \in \Sigma^*$ there is a model $e \in \mathrm{Cons}(R)$ with φ_e total and $\varphi_e(x) = y$. Consequently $\mathrm{Cons}(R)$ contains an infinite family of total models any two of which differ at every $x \notin \mathrm{dom}(R)$.

(b) For every $e \in \mathrm{Cons}(R)$ there are infinitely many $e' \neq e$ with $\varphi_{e'} = \varphi_e$; all of them lie in $\mathrm{Cons}(R)$.

Step	Statement	Status	Level
Known	To know that a theory is final is for the record to rule out every computable rival (definition 2.6)	definition	—
P1	Every use yields a finite record	premise	—
P2	Every usable system, every method included, is computable or a computable stochastic process	premise	—
T1	Every record has infinitely many consistent models, deterministic or stochastic, that diverge on unobserved settings (theorem 3.1)	theorem	programs, extensions, chances
K1	If a method certifies a model at a record, some computable world producing that record makes the certification wrong (corollary 3.2)	corollary	extensions, chances
T2	For every deterministic computable method and every record, some computable world consistent with the record defeats it; a randomized one succeeds with no fixed positive probability (theorem 4.1; for active methods, remark 4.3; at any oracle level, theorem 4.5)	theorem	extensions
P2*	Every usable method faces all worlds computable at its own level (premise P2*)	premise, weaker than P2	—
L	Every consistent method is a bias that the record does not determine (lemma 4.6)	lemma	—
T3	Which consistent model is shortest depends on the reference machine: a near-tie of δ bits is reversed by a change of machine that moves no description length by more than δ, and rivals within about $K(x)$ bits of the shortest disagree at any unobserved setting x (theorems 5.3 and 5.5, proposition 5.7, and corollary 5.8)	theorem	extensions
T5	No method separates formulations with identical predictions, or mutually absolutely continuous chance hypotheses with probability 1 under both (theorem 6.1)	theorem	programs, chances
T4	The universal predictor is within $K(\nu)$ bits of every model ν; no computable predictor is (theorem 7.1 and proposition 7.2)	theorem	extensions, chances
C1	No method of any kind can certify a theory, deterministic or stochastic, as final	from T1, K1	
C2	No usable method is guaranteed, even with any fixed probability, to settle on a final theory	from P2 (P2* for deterministic methods), T2, L	
C3	Formulations with identical predictions, and chance hypotheses that agree on which events have probability zero, are never told apart with certainty	from T5 (T1(b) illustrates)	
C′	Simplicity narrows the field but cannot decide among near-tied survivors that disagree about the unobserved; choosing among them is an imposed constraint; compression still predicts	from L, T3, T4; P2 to read T4's benchmark as unusable	

Table 1: The steps of the argument in section order. "Level" says whether a step concerns programs (indices), extensions (behavior) or chance hypotheses (measures). With definition 2.6, C1 is the main claim that no theory can be known to be final.

(c) Let $r \in \Sigma^*$ be the outcome string of any finite record, functional or not (definition 2.2). There are infinitely many computable probability measures ν on Σ^∞ with $\nu(r) > 0$ whose predictions after r pairwise differ: $\nu(0 \mid r) \neq \nu'(0 \mid r)$ for $\nu \neq \nu'$, where $\nu(a \mid r) = \nu(ra)/\nu(r)$.

Proof. (a) Since $\mathrm{dom}(R)$ is finite, the function

$$f_{x,y}(z) = \begin{cases} y' & \text{if } (z, y') \in R, \\ y & \text{if } z = x, \\ \varepsilon & \text{otherwise} \end{cases}$$

differs from the constant function $z \mapsto \varepsilon$ on finitely many arguments, so it is total computable; any index e for it lies in $\mathrm{Cons}(R)$ and has $\varphi_e(x) = y$. For the stronger family, let $g_y(z)$ be y' if $(z, y') \in R$ and y otherwise. Each g_y is total computable and consistent with R, and for $y \neq y'$ the models $g_y, g_{y'}$ differ at every $z \notin \mathrm{dom}(R)$.

(b) This is the padding lemma for acceptable numberings: there is a computable injection $p : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ with $\varphi_{p(e,n)} = \varphi_e$ for all e, n (Rogers, 1967, ch. 1). Membership in $\mathrm{Cons}(R)$ depends only on φ_e.

(c) Let $k = |\Sigma|$ and $m = |r|$. For each rational $q \in (0, 1)$ let ν_q make the first m symbols independent and uniform on Σ, and every later symbol independently 0 with probability q and each other symbol with probability $(1 - q)/(k - 1)$. Each ν_q is computable, $\nu_q(r) = k^{-m} > 0$, and $\nu_q(0 \mid r) = q$. □

The construction is folklore: Kukla's recipe for a rival to any theory T—the theory T'' that agrees with T wherever the world is observed and with some other theory elsewhere (Kukla, 1993)—is the family g_y made precise. Part (b) gives models that agree everywhere and differ only as programs; we return to it in section 6.

Corollary 3.2 (No certification)**.** *Let M be any method whatever: computable or not, deterministic or randomized.*

(a) If at a finite functional record R the method M certifies a deterministic model e as final (definition 2.4) with positive probability, then there is a total computable world f consistent with R, with $\varphi_e \neq f$, in which M receives R and certifies e with the same probability.

(b) If at a record with outcome string r the method M certifies a stochastic model i as final with positive probability, then there is a computable measure $\mu \neq \nu_i$ with $\mu(r) > 0$; in the world μ, M certifies i wrongly with positive probability.

Hence every method that certifies at some record is wrong on some computable world that produces that record. A certification can be correct only relative to a restricted class of worlds, one that excludes the rivals of theorem 3.1.

Proof. (a) Pick $x \notin \mathrm{dom}(R)$. If $\varphi_e(x)$ is undefined, every total g_y of theorem 3.1(a) is a world consistent with R in which e is not final. Otherwise choose $y \neq \varphi_e(x)$; the world g_y is consistent with R and differs from φ_e at x. In either case the world produces the record R, and M's output depends only on its input (and, if M is randomized, on its own random bits), so M certifies e there with the same probability, wrongly.

(b) Take $\mu = \nu_q$ from the proof of theorem 3.1(c), with $q \neq \nu_i(0 \mid r)$ if $\nu_i(r) > 0$ and q arbitrary if $\nu_i(r) = 0$. Then $\mu(r) > 0$ and $\mu \neq \nu_i$. In the world μ the outcome string r occurs

with probability $\mu(r) > 0$, and on receiving it M certifies i with the same positive probability as before, wrongly. □

The restriction to a class of worlds is a bias in the sense of definition 2.7: it is not supplied by the record.

Remark 3.3 (Each rival is refuted eventually). Every model that gives a different output from a world f at some setting is refuted after finitely many observations of the canonical presentation of f. (A model that is undefined at some setting leaves Cons once that setting is observed, though no finite computation exhibits its failure.) So each rival constructed in theorem 3.1(a) is only transiently consistent with the growing record, and one might hope that a method could at least *converge* on a final theory without ever certifying it. Section 4 shows that no usable method can guarantee even this. The rivals that are never refuted—models that agree on every possible record—are the subject of section 6.

4 No usable method finds one

Corollary 3.2 rules out certifying a final theory. A weaker goal remains: a method might converge on a final theory without ever announcing that it has. This section shows that no usable method is guaranteed, even with any fixed probability, to settle on a final theory, and that whatever success a method has is bought by a bias the record does not supply.

Fix the length-lexicographic enumeration $x_0, x_1, x_2, \ldots$ of Σ^*, a computable bijection $\mathbb{N} \to \Sigma^*$, and identify a total $f : \Sigma^* \to \Sigma^*$ with the sequence $f(x_0), f(x_1), \ldots$. For a finite sequence $\sigma = (y_0, \ldots, y_{n-1})$ write $R_\sigma = \big((x_0, y_0), \ldots, (x_{n-1}, y_{n-1})\big)$, write $\sigma\tau$ for concatenation, $\sigma \sqsubseteq \tau$ for the prefix order, and $f \restriction n = (f(x_0), \ldots, f(x_{n-1}))$. For a record R already in hand, $R \frown R_\tau$ is the record R followed by the observations of R_τ; the presentation of f after R is the chain $R \frown R_{f\restriction 0} \sqsubseteq R \frown R_{f\restriction 1} \sqsubseteq \cdots$ (definition 2.5). Fix two distinct symbols $0, 1 \in \Sigma$.

Any finite functional record R determines a finite sequence σ_R: take the least n with $\mathrm{dom}(R) \subseteq \{x_0, \ldots, x_{n-1}\}$ and set the ith entry to the outcome R records at x_i, or to 0 if it records none. Every total f extending σ_R is consistent with R.

Theorem 4.1 (T2: Non-identifiability from any record; after Putnam (1963); Gold (1967))**.** *Let L be a computable method (definition 2.5), possibly partial, and R a finite functional record. There is a total computable world f, consistent with R, that L does not identify in the limit after R.*

Proof. We build finite sequences $\sigma_0 \sqsubseteq \sigma_1 \sqsubseteq \cdots$ with $\sigma_0 = \sigma_R$. Given σ_s, search, dovetailing over k and computation time, for a k such that $h = L(R \frown R_{\sigma_s 0^k})$ is defined and

$$\varphi_h\big(x_{|\sigma_s|+k}\big)\!\downarrow = 0,$$

that is, the current conjecture predicts that the record continues with 0. If the search succeeds, set $\sigma_{s+1} = \sigma_s 0^k 1$.

Suppose the search at some stage s never succeeds. Then L does not identify $g = \sigma_s 0^\infty$ after R. If it did, it would be defined on every record of the presentation of g after R, and from some point on its conjecture would be a fixed index for g; that index outputs 0 at every argument beyond σ_s, so the search would succeed. The world g is total computable and extends σ_R, so it is consistent with R.

Otherwise every stage succeeds, and $f = \bigcup_s \sigma_s$ is total computable and extends σ_R, so it is consistent with R. The conjecture h_s found at stage s is the output of L on a record of the presentation of f after R, since $\sigma_s 0^k \sqsubseteq f$, and it predicts 0 where f has 1, so $\varphi_{h_s} \neq f$. These records grow with s. If L converged on f to some h^*, then $h_s = h^*$ for all large s, so $\varphi_{h^*} \neq f$. $\square$

A method that does well on the simplest worlds (here, those that are eventually constant) for that very reason never settles on the right theory in another computable world, and that world can be chosen to agree with any record already in hand. The world is not exotic. The construction is uniform in L and R. If every stage succeeds, $K(f) \leq K(L, R) + O(1) \leq K(L) + K(R) + O(1)$, where $K(L)$ is the complexity of an index for L. If stage s stalls, the world is eventually constant and $K(g) \leq K(L, R) + K(s) + O(1)$. Either way its complexity is set by the method, the record and at most one stage number, and its records are compressible, not atypical or non-stochastic in the sense of algorithmic statistics (Gács et al., 2001).

Remark 4.2 (Randomized methods). A method may also flip a coin of computable bias. The theorem covers such methods in the following form. A probabilistic method that identifies every computable world with probability greater than $1/2$ can be replaced by a deterministic computable method that identifies the same worlds (Pitt, 1989, Thm. 3.21 with $n = 1$), so by theorem 4.1 no probabilistic method identifies every computable world with probability greater than $1/2$. Success with a smaller probability, bounded below by a fixed $p > 0$ for every world, corresponds to identification by a finite team of deterministic methods, and the class of all computable worlds is not identifiable by any finite team either (Pitt, 1989, Thm. 3.36; Smith, 1982, Thm. 3.1; Pitt and Smith, 1988). Both facts hold relative to any record: the class of computable worlds extending σ_R is not finite-team identifiable, since a team for it would yield one for all computable worlds via $g \mapsto \sigma_R g$ and the corresponding computable shift of indices. So, given any record, no usable randomized method that receives the presentation identifies every computable world consistent with it with any fixed positive probability. This is a uniform statement: it does not say that the method fails almost surely in some single world, and a randomized method can identify every computable world with *some* positive probability, for instance by guessing an index at random and keeping it.

Remark 4.3 (Active methods). A method may also choose which settings to observe. An *active* method reads the record so far, and possibly random bits, and outputs a conjecture and a next setting to query; the world answers the query, and the answer extends the record. Active methods gain nothing here. Given a computable active method M and a record R, let L_M be the method that receives the presentation after R and, on the record $R \frown R_{f\restriction n}$, simulates M from R for n computation steps with the same random bits, answering each query from the record and halting the simulation at the first query the record does not yet answer; L_M outputs the last conjecture of M reached. L_M is computable, and randomized exactly when M is. Every setting is eventually answered, so every step of a run of M in which each step halts is eventually reached; if on given random bits M identifies f, its conjectures are eventually constant at an index of f, and so are those of L_M. Hence L_M identifies every world that M identifies, with at least the same probability. Applied to L_M, theorem 4.1 and remark 4.2 give, for every record, a computable world consistent with it that M does not identify, or, for randomized M and every $p > 0$, one that M identifies with probability less than p.

Corollary 4.4 (No record rules out failure)**.** *Let L be a usable method, possibly partial, and R a finite functional record that L has observed; L may receive the presentation or choose its own queries. If L is deterministic, some computable world consistent with R defeats it: in that*

world, L never settles on the right theory. If L is randomized, then for every $p > 0$ there is a computable world consistent with R that L identifies with probability less than p. In either case no record tells L's user that the world is not such a world.

Proof. By premise P2 L is computable or a computable probabilistic method; apply theorem 4.1 or remark 4.2, and for active methods remark 4.3, to R. □

The limit does not depend on the level of computing power. Theorem 4.1 is not an artifact of ordinary computers. Fix any set $A \subseteq \mathbb{N}$ as an oracle, and let $(\varphi_e^A)_{e\in\mathbb{N}}$ be an acceptable numbering of the functions partial computable relative to A, which exists by the relativized universal-function and *s-m-n* theorems (Rogers, 1967, ch. 9); the proof below uses only that the universal function is A-computable. Call a model at level A an index e of φ_e^A, and a world at level A a total A-computable function; identification at level A is defined as in definition 2.5, with φ_e^A in place of φ_e.

Theorem 4.5 (T2 at every level)**.** *Let $A \subseteq \mathbb{N}$, let L be an A-computable method, possibly partial, that outputs models at level A, that is, indices in (φ_e^A) rather than ordinary indices, and let R be a finite functional record. There is a total A-computable world f, consistent with R, that L does not identify at level A after R.*

Proof. The proof of theorem 4.1 goes through with φ^A in place of φ. Given σ_s (with $\sigma_0 = \sigma_R$), search for k such that $h = L(R \frown R_{\sigma_s 0^k})$ is defined and $\varphi_h^A(x_{|\sigma_s|+k})\!\downarrow = 0$. The computations of L and of φ_h^A can be simulated with access to A, so the search, dovetailed over k and computation time, is A-effective. If it never succeeds at stage s, then L does not identify $\sigma_s 0^\infty$ at level A, for the same reason as before, and that world is computable, hence A-computable. Otherwise set $\sigma_{s+1} = \sigma_s 0^k 1$. Then $f = \bigcup_s \sigma_s$ is total, extends σ_R, and is A-computable, since every stage is effective relative to A. Each conjecture h_s found at stage s predicts 0 where f has 1, so if L converged on f to some h^* then $\varphi_{h^*}^A \neq f$. □

The relativized theorem needs the worlds to be at the method's level: a method strictly more powerful than the worlds it faces can succeed, as the ideal-agent concession below shows. For deterministic methods, C2 therefore needs only the following premise.

Premise P2* (Methods and worlds share a level)**.** For every usable method L there is a set $A \subseteq \mathbb{N}$ such that L is A-computable, its outputs are read as models at level A, and its guarantee must cover every total A-computable world consistent with the record: every such world is a possible world. A model at level A is final for a world f if it computes f relative to A.

Refuted by: a usable method more powerful than every world its guarantee must cover, for instance a usable oracle whose power the theories it chooses among cannot share.

Premise P2, together with the guarantee of definition 2.5, which covers every total computable world consistent with the record, is the case $A = \emptyset$ of P2*. Under P2*, theorem 4.5 applies at the method's own level, so a usable hypercomputer of oracle strength, whose power the theories share, does not escape C2. Hypercomputers not equivalent to a single oracle, such as infinite-time Turing machines, would need their own argument. The reductions of Pitt (1989) behind remark 4.2 are uniform simulations and should relativize, but we have not verified this, so P2* serves for deterministic methods only.

Lemma 4.6 (L: Every consistent method is a bias)**.** *Let L be a method that outputs, on each functional record R, a model in* $\mathrm{Cons}(R)$.

(i) L is a record-indexed constraint (definition 2.7): with $c_R(e) = 0$ if $e = L(R)$ and $c_R(e) = 1$ otherwise, $\mathrm{Sel}_c(R) = \{L(R)\}$. *For every functional record R there is a method L' that outputs consistent models, is computable if L is, and selects differently from L at R.*

(ii) For a finite functional record R, let K_L be the class of total computable worlds consistent with R that L identifies after R. If L is computable (deterministic), some world consistent with R lies outside K_L; and if L identifies after R the eventually-zero worlds consistent with R, some world lies inside K_L.

Proof. (i) The first claim is immediate from the definition, since $L(R) \in \mathrm{Cons}(R)$ has cost 0 and every other model cost 1. For the second, $\mathrm{Cons}(R)$ is infinite by theorem 3.1, so it contains some $e' \neq L(R)$. Let L' agree with L on every record other than R, and let $L'(R) = e'$. Then L' outputs consistent models, it differs from L only on one input, so it is computable if L is, and $\mathrm{Sel}_{c'}(R) = \{e'\} \neq \{L(R)\}$, where c' is the constraint of L' given by (i). So the record does not determine the constraint, which is what it means for the constraint to be imposed (definition 2.7). For (ii), the first claim is theorem 4.1; for the second, the world $\sigma_R 0^\infty$ is eventually zero and consistent with R. □

So choosing a method is choosing a class K_L of worlds to be right about, and no record decides whether the actual world belongs to that class. Section 5 examines simplicity as a way of making this choice.

The ideal-agent concession. Theorem 4.1 binds computable methods, and the restriction is not idle. A method with an oracle for the totality problem can output the least total index consistent with the current record; every smaller total index computes a different function and is eventually refuted, so this *identification by enumeration* converges to a correct model on every total computable world (Putnam, 1963; Gold, 1967). An ideal agent of this kind therefore does converge on the true behavior of any deterministic computable world, though it never knows when it has. The limit of theorem 4.1 is computational.

5 Simplicity does not single one out

The most common preference in the search for a theory of everything is simplicity: among the candidates that fit the data, prefer the shortest. Which candidates fit is fixed by the record. Which of them is shortest depends on the reference machine, the universal computer in which lengths are measured, and there are always rivals that fit equally well, are nearly as short, and disagree about what has not been observed.

5.1 Description length as a constraint

Definition 5.1 (Prefix description system)**.** A *prefix description system* is a partial computable function $U : \{0,1\}^* \times \Sigma^* \rightharpoonup \Sigma^*$ whose set of programs $D_U = \{p : U(p, x)\downarrow \text{ for some } x\}$ is prefix-free. A program $p \in D_U$ determines the model $x \mapsto U(p, x)$; by the *s-m-n* theorem an index e_p with $\varphi_{e_p} = U(p, \cdot)$ is computable from p. U is *universal* if for every prefix description system V there is a string σ_V with $U(\sigma_V q, x) \simeq V(q, x)$ for all q, x. Universal systems exist (Chaitin, 1975; Li and Vitányi, 2019).

Definition 5.2 (Description length of a record)**.** For a universal U and a nonempty finite functional record R, a program $p \in D_U$ is *consistent with* R if $U(p, x) = y$ for all $(x, y) \in R$. Let

$$K_U(R) = \min\{|p| : p \in D_U \text{ consistent with } R\},$$

and let the *MDL selection relative to* U be the set of models e_p for which p is consistent with R and $|p| = K_U(R)$. For a string s we write $K_U(s) = K_U(\{(\varepsilon, s)\})$; K with no subscript denotes K_U for a fixed reference U.

$K_U(R)$ is finite, because a lookup table for R is a consistent program. Minimum description length (MDL) is a constraint in the sense of definition 2.7: let $c_U(e) = \min\{|p| : p \in D_U,\ e_p = e\}$, with $c_U(e) = \infty$ if there is no such p. Since p is consistent with R exactly when $e_p \in \mathrm{Cons}(R)$, $\mathrm{Sel}_{c_U}(R)$ is the MDL selection relative to U. It is *imposed* in the sense of definition 2.7: by theorem 5.3(i) below, the same record yields different MDL selections under different universal machines.

5.2 The choice of machine

Theorem 5.3 (T3: MDL is machine-relative)**.** *Let U be universal, R a nonempty finite functional record, and $e \in \mathrm{Cons}(R)$. There is a universal prefix description system U' such that*

(i) for every record $R' \sqsupseteq R$ with $e \in \mathrm{Cons}(R')$, the MDL selection relative to U' has a unique shortest consistent program, and it computes φ_e;

(ii) $K_{U'}(R') \le K_U(R') + 1$ for every nonempty functional record R';

(iii) $K_U(R') \le K_{U'}(R') + K_U(e) + O(1)$ for every nonempty functional record R', where $K_U(e)$ is the length of a shortest U-program that outputs e and the $O(1)$ depends only on U.

Proof. Define $U'(0, x) \simeq \varphi_e(x)$ and $U'(1p, x) \simeq U(p, x)$, with U' undefined on all other programs. Then $D_{U'} \subseteq \{0\} \cup 1D_U$, which is prefix-free: 0 is incomparable with every string beginning with 1, and $1p \sqsubseteq 1q$ forces $p \sqsubseteq q$, hence $p = q$ for $p, q \in D_U$. U' is universal with $\sigma'_V = 1\sigma_V$.

(i) Since $e \in \mathrm{Cons}(R')$ and $R' \neq \emptyset$, φ_e halts on some input, so $0 \in D_{U'}$ and 0 is consistent with R'. The empty string is not in $D_{U'}$, and $1 \in D_{U'}$ only if $\varepsilon \in D_U$, which prefix-freeness and universality exclude. So 0 is the unique shortest consistent program.

(ii) Each U-program p becomes the U'-program $1p$.

(iii) Let V read a U-program a (self-delimiting, since $a \in D_U$), decode $e = U(a, \varepsilon)$, then read a U'_e-program q and simulate $U'_e(q, \cdot)$, where U'_e is the machine U' built from e; the simulation is uniform in e. Each U'-program q then has the U-program $\sigma_V a q$, where a is a shortest U-program for e, so $|a| = K_U(e)$. □

The invariance theorem (Kolmogorov, 1965, §3; Li and Vitányi, 2019) says that two universal machines assign description lengths within an additive constant of each other. Theorem 5.3 says what that constant means on finite data. One extra bit in the reference machine (ii) makes any chosen consistent candidate the unique simplest one, and it stays so until the record refutes it (i). The price, measured in the original machine, is at most about $K_U(e)$, the length of the candidate being favored (iii). Promoting a candidate far from the minimum, such as a lookup table for the record, needs a change of machine with large translation constants (the converse after theorem 5.5). Promoting a near-tie needs only a small one, in the following sense.

Definition 5.4 (Pointwise cost of a change of machine)**.** Universal prefix description systems U, U' differ *pointwise by at most* c if $|K_U(R') - K_{U'}(R')| \le c$ for every nonempty finite functional record R'.

The pointwise cost bounds how far a change of machine moves any description length. It is not the translation constant of the invariance theorem, the length of a prefix σ with $U(\sigma q, x) \simeq U'(q, x)$ for all q, x; that constant bounds the pointwise cost from one side, but it need not be small.

Theorem 5.5 (T3, gap form: near-ties are machine-relative)**.** *Let U be universal, R a nonempty finite functional record, $\delta \ge 0$ an integer, and $q \in D_U$ a program consistent with R with $|q| \le K_U(R) + \delta$. There is a universal prefix description system U' such that*

(i) for every record $R' \sqsupseteq R$ with which q is consistent, the unique shortest U'-program consistent with R' computes the model $x \mapsto U(q, x)$;

(ii) $K_{U'}(R') \le K_U(R') + 1$ and $K_U(R') \le K_{U'}(R') + \delta$ for every nonempty finite functional record R', so U and U' differ pointwise by at most $\max(\delta, 1)$.

Proof. Let $\ell = \max(0, |q| - \delta - 1)$ and $t = 0^\ell$. Define $U'(0s, x) \simeq U(s, x)$ for every s and $U'(1t, x) \simeq U(q, x)$, with U' undefined on all other programs. Then $D_{U'} \subseteq 0D_U \cup \{1t\}$, which is prefix-free: $1t$ is incomparable with every string beginning with 0, and $0s \sqsubseteq 0s'$ forces $s = s'$ for $s, s' \in D_U$. U' is universal with $\sigma'_V = 0\sigma_V$. Every U'-program other than $1t$ has the form $0s$ with $s \in D_U$ and computes the same model as s, so

$$K_{U'}(R') = \begin{cases} \min\big(K_U(R') + 1,\ \ell + 1\big) & \text{if } q \text{ is consistent with } R', \\ K_U(R') + 1 & \text{otherwise.} \end{cases} \tag{1}$$

Note that $K_U(R') \ge 1$ for every nonempty R': the empty program is not in D_U, since otherwise $D_U = \{\varepsilon\}$ and U would not be universal.

(i) Let $R' \sqsupseteq R$ with q consistent with R'. A program consistent with R' is consistent with R, so $K_U(R') \ge K_U(R) \ge |q| - \delta$. Hence $\ell + 1 \le \max(1, |q| - \delta) \le K_U(R') < K_U(R') + 1$. So $1t$, which is consistent with R', is strictly shorter than every consistent program $0s$, and it computes $x \mapsto U(q, x)$.

(ii) The first inequality is immediate from (1). For the second, if $K_{U'}(R') = K_U(R') + 1$ there is nothing to prove. Otherwise $K_{U'}(R') = \ell + 1$ and q is consistent with R', so $K_U(R') \le |q| \le \ell + 1 + \delta$. □

Two features of the theorem matter. First, the cost is pointwise only. The proof gives no simulation of U' in U shorter than about $K_U(q)$ bits, since U' must be told q; so the translation constant may be as large as the bound of theorem 5.3(iii), while no description length moves by more than $\max(\delta, 1)$. Second, the converse holds for translation constants. If $U(\sigma q', x) \simeq U'(q', x)$ for all q', x and $K_{U'} \le K_U + c'$, then the shortest U'-program consistent with R, prefixed by σ, is a U-program for the same model, so the U'-selection lies within $|\sigma| + c'$ bits of the U-minimum. A change of machine with small translation constants reorders only near-ties.

Remark 5.6 ("Unnatural" machines). The standard reply is that U' is unnatural. But naturalness of the reference machine is itself an imposed constraint, and no known principle privileges one universal machine over another. The most direct attempt, a stationary distribution for the process of universal machines simulating one another, does not exist on the set of all universal machines; it exists only on restricted subclasses, which reintroduces the choice (Müller, 2010).

5.3 Near-minimal rivals

Proposition 5.7 (Near-minimal rivals)**.** *Let U be universal, R a nonempty finite functional record, and p a shortest program consistent with R. For every $x \notin \mathrm{dom}(R)$ and every $y \in \Sigma^*$ there is a program q consistent with R such that $U(q, x) = y$ and*

$$|q| \leq K_U(R) + K(x, y) + O(1),$$

where the $O(1)$ depends only on U. In particular, for every such x there is a consistent q that disagrees at x with the shortest program p (that is, $U(q, x) \neq U(p, x)$, including the case $U(p, x)\uparrow$) and $|q| \leq K_U(R) + K(x) + O(1)$.

Proof. Let V be the prefix description system that, on program s and input z, finds the unique prefix a of s in D_U (by dovetailing U on the prefixes of s), requires $U(a, \varepsilon) = \langle x, y\rangle$, confirms that the remainder p of s lies in D_U, and then outputs y if $z = x$ and $U(p, z)$ otherwise. $D_V \subseteq \{ap : a, p \in D_U\}$, which is prefix-free. Take a a shortest U-program for $\langle x, y\rangle$ and set $q = \sigma_V ap$. For $(x', y') \in R$ we have $x' \neq x$, so $U(q, x') = U(p, x') = y'$; and $U(q, x) = y$. For the last claim, since $|\Sigma| \geq 2$, choose a single symbol y different from $U(p, x)$ (any symbol if $U(p, x)\uparrow$); then $K(x, y) \leq K(x) + O(1)$. □

Corollary 5.8 (Near-minimal rivals can be promoted)**.** *Let U, R and p be as in proposition 5.7, and $x \notin \mathrm{dom}(R)$. There is a universal U' that differs from U pointwise by at most $K(x) + O(1)$ and whose unique shortest program consistent with R computes a model that disagrees with p at x.*

Proof. Take the rival q of proposition 5.7 that disagrees with p at x, and apply theorem 5.5 with $\delta = |q| - K_U(R) \leq K(x) + O(1)$. □

Two margins are in play here, and they are different. The first is the margin of a *change of machine*: by corollary 5.8, a change that moves no description length by more than $K(x) + O(1)$ bits reverses which candidate MDL selects, in favor of one that predicts differently at x. For a setting x of bounded complexity this margin is a constant. For the canonical record of length n every unobserved setting is some x_m with $m \geq n$; the next one costs $K(x_n) \leq K(n) + O(1)$, at most about $\log n$ bits, and some later ones, such as x_{2^k} with $2^k \geq n$, cost only about $\log \log n$.

The second margin is that of *algorithmic statistics* for a fixed machine. Algorithmic statistics draws the most that can be drawn from a record without a choice among near-ties. The Kolmogorov structure function and the minimal sufficient statistic split the record's information into a meaningful part, a simplest model for which the record is typical, and an accidental remainder (Gács et al., 2001; Vereshchagin and Vitányi, 2004). Three features of that split leave the near-ties standing. It is determined only up to additive terms logarithmic in the length n of the record; its models are sets or distributions that contain the record, so it describes what has been observed, not how the record extends; and it is neither computable nor usefully approximable (Vereshchagin and Vitányi, 2004, Sec. VII). The rivals of proposition 5.7 at the next setting cost at most $K(n) + O(1)$, about $\log n$, extra bits, so they lie inside the margin within which the structure function, for any fixed machine, declares itself indifferent. This margin does not depend on a change of machine.

5.4 Other preferences, and serious rivals

Simplicity is not the only preference at work in the search for a final theory. Naturalness, beauty, symmetry, and unification are also orderings on candidate models, and so constraints in the sense of definition 2.7: each ranks models that the record cannot tell apart.

Nor are the surviving rivals only gerrymanders. Physics supplies rivals of comparable compactness that agree on every prediction. When two formulations are related by a computable, prediction-preserving translation τ, their description lengths differ by at most $K(\tau) + O(1)$: a program for one formulation, prefixed by a self-delimiting program for τ, is a program for the other. For the explicit translations below $K(\tau)$ is the length of a short procedure, so if one formulation is a shortest consistent program, the other is a near-tie of at most $K(\tau)+O(1)$ bits, and by theorem 5.5 a change of machine that moves no description length by more than that makes it the unique selection. Newtonian, Lagrangian and Hamiltonian mechanics agree, for conservative systems with a regular Lagrangian, through the Legendre transform, while their primitives differ: forces along trajectories, a variational principle over histories, flows on phase space. Heisenberg's matrix mechanics and Schrödinger's wave mechanics began from incompatible pictures and were shown equivalent, incompletely by Schrödinger (Schrödinger, 1926) and rigorously by von Neumann (von Neumann, 1932); string-theoretic dualities are candidates of the same kind (section 6).

6 What even an ideal agent cannot know

An agent with unlimited computing power escapes theorem 4.1. In a deterministic computable world every divergent rival is refuted after finitely many observations (remark 3.3), and enumeration with an oracle for totality converges on the true behavior (section 4). Two things remain even for such an agent: rival *formulations* of one final theory, and rival *chance hypotheses*, which leave the final theory itself undetermined.

Theorem 6.1 (T5: Permanent underdetermination)**.**

(a) Formulations. If two models are extensionally equal (definition 2.2)—deterministic models e, e' with $\varphi_e = \varphi_{e'}$, or stochastic models i, j with $\nu_i = \nu_j$—then they assign the same outcome, or the same probability, to every possible record. No method of any kind, computable or not, separates them.

(b) Chance hypotheses. Let μ and μ' be distinct computable measures on Σ^∞ that are mutually absolutely continuous, that is, agree on which events have probability zero: $\mu(B) = 0$ if and only if $\mu'(B) = 0$, for every Borel set B. Let L be any method, computable or not, mapping finite outcome strings to models. If L converges to a model of μ with μ-probability 1, then it converges to a model of μ' with μ'-probability 0. In particular, no method identifies the correct one of μ, μ' with probability 1 under both.

Proof. (a) A record's outcomes are fixed by its settings and the model's behavior on them (definition 2.2); extensionally equal models agree on both, so every method receives the same input whichever of them is true.

(b) For a set I of models (a countable set of indices), the event that L converges to a member of I is

$$A_I = \bigcup_N \bigcup_{i \in I} \bigcap_{n \geq N} \{\omega \in \Sigma^\infty : L(\omega_{\leq n}) = i\}.$$

Each set $\{\omega : L(\omega_{\le n}) = i\}$ depends only on the first n symbols, so it is a finite union of cylinders; the unions and intersections are countable. Hence A_I is Borel whatever L is. Let I and I' be the sets of models of μ and of μ', that is, of indices i with $\nu_i = \mu$ and with $\nu_i = \mu'$. They are disjoint because $\mu \neq \mu'$, so $A_I \cap A_{I'} = \emptyset$. If $\mu(A_I) = 1$ then $\mu(A_{I'}) = 0$, and by absolute continuity $\mu'(A_{I'}) = 0$. □

The argument uses no property of μ, μ' beyond mutual absolute continuity. The restriction to computable measures loses nothing: a probability measure that has a stochastic model is lower semicomputable, and a lower semicomputable probability measure is computable, since $\nu(r) = 1 - \sum_{r' \neq r,\, |r'|=|r|} \nu(r')$ is also upper semicomputable (Li and Vitányi, 2019).

A method that uses internal randomness is covered as well, provided each conjecture depends on $\omega_{\le n}$ and on finitely many of the method's random bits ρ (possibly a number of them that depends on the bits read). Then the success event B is a Borel subset of $\Sigma^\infty \times \{0,1\}^\infty$, and (b) applies to the products $\mu \times \lambda$ and $\mu' \times \lambda$, where λ is the uniform measure on the random bits. These products are again mutually absolutely continuous: by Fubini's theorem, $(\mu \times \lambda)(B) = \int \mu(B^\rho)\, d\lambda(\rho)$, which vanishes if and only if $\mu(B^\rho) = 0$ for λ-almost every ρ, hence if and only if $\mu'(B^\rho) = 0$ for λ-almost every ρ, hence if and only if $(\mu' \times \lambda)(B) = 0$.

Instances of (a). The trivial instance, as an illustration only, is the padding lemma: every behavior has infinitely many programs (theorem 3.1(b)). The serious instances are rival *formulations* that physicists take seriously.

- **Dualities.** Pairs of theories with very different ingredients are held to describe the same physics. In T-duality, string theory on a circle of radius R is equivalent to string theory on a circle of radius α'/R (Polchinski, 1998). In the AdS/CFT correspondence, string theory on an anti-de Sitter space is conjectured to be equivalent to a gauge theory without gravity on its boundary (Maldacena, 1998). Where such a duality holds, no record says which side is fundamental, or whether the space has one size or the other.

- **Interpretations of quantum mechanics.** The following all assign Born-rule probabilities to every record:

  - Copenhagen-style quantum mechanics, in which the probabilities are fundamental;
  - de Broglie–Bohm theory in quantum equilibrium, which is deterministic, with randomness confined to the initial positions (Bohm, 1952; Dürr et al., 1992);
  - Everettian quantum mechanics, which is deterministic unitary evolution, with Born weights for branches (Everett, 1957; Wallace, 2012).

  As stochastic models they are therefore extensionally equal. The question "is the world fundamentally random?" is not settled by any record.

Two candidates do not belong on the list. Objective-collapse theories (Ghirardi et al., 1986) predict small departures from standard unitary quantum predictions (suppressed macroscopic interference, spontaneous heating), so they are rivals that records can in principle refute, not instances of (a). Superdeterministic proposals are programmatic: their proponents describe the models as "still in their infancy" (Hossenfelder and Palmer, 2020, Sec. 5).

Instances of (b). The simplest example is two coin models. Under μ every toss lands heads with probability $1/2$. Under μ' the first toss lands heads with probability $1/3$ and every later toss with probability $1/2$. These are distinct chance hypotheses, since they disagree about a definite event. Because they differ in only finitely many conditional probabilities, each of which is strictly between 0 and 1, they are mutually absolutely continuous (the density of μ' with respect to μ is $2/3$ or $4/3$). More generally, two theories of chance are mutually absolutely continuous if their conditional probabilities differ at only finitely many histories in total, at each of which both conditionals lie in $(0, 1)$: the density is then a finite product of ratios bounded away from 0 and ∞. For independent trials with pairwise equivalent factors μ_n, μ'_n, Kakutani's criterion gives mutual absolute continuity exactly when $\prod_n \int \sqrt{d\mu_n \, d\mu'_n} > 0$; for Bernoulli factors with probabilities p_n, q_n bounded away from 0 and 1, this holds if and only if $\sum_n (p_n - q_n)^2 < \infty$ (Kakutani, 1948). A method may still guess well: if the first toss has heads probability 0.99 under one hypothesis and 0.01 under the other, guessing from that toss is right with probability 0.99 either way. What no method achieves is convergence on the correct hypothesis with probability 1 under both. Positive results exist for restricted classes. Vitányi and Chater (2017) identify a computable measure in the limit when the data are independent draws from a computably enumerable class of measures; for dependent data their learner returns a measure for which the sequence is typical, and there may be several, which is the gap (b) exhibits. Without such restrictions, algorithmic identification of probabilities fails (Bienvenu et al., 2018).

Scope. The theorem takes a physical formulation to be a model in the sense of definition 2.2, so that whether a formulation is fundamental is a property of the model rather than of its predictions. Part (a) then shows that the *fundamental formulation* of a theory of everything is never settled by data: two candidates with the same predictions may disagree about what exists (strings in one space or a gauge theory on its boundary; particles with trajectories or a branching wave function). Part (b) reaches the final theory itself: for theories of chance, a rival that agrees on which events have probability zero puts the final theory, in the sense of definition 2.4, beyond identification with probability 1 under both.

7 What the search is good for

The preceding sections are negative about *finality*. Here we explain why the search for compact, unified theories is still worthwhile. The reason is prediction. We first state the strongest predictive guarantee available, then show that no usable method has it, and finally say what compression offers in its place.

The ideal benchmark. Take stochastic models as in definition 2.2(b), with the enumeration $\nu_1, \nu_2, \ldots$ fixed there and the outcomes of a record forming a string $z = z_1 \cdots z_n$, and write $K(\nu) = \min\{K(i) : \nu_i = \nu\}$ for the least prefix complexity of an index of ν. The universal mixture

$$\mathbf{M}(z) = \sum_{i \geq 1} 2^{-K(i)} \nu_i(z)$$

weights every model by its compressed length and predicts with $\mathbf{M}(a \mid z) = \mathbf{M}(za)/\mathbf{M}(z)$. Up to a multiplicative constant it is Solomonoff's a priori probability (Solomonoff, 1978; Li and Vitányi, 2019).

Theorem 7.1 (Relative optimality)**.** *For every lower semicomputable semimeasure ν and every string $z = z_1 \cdots z_n$,*

$$\sum_{t=1}^{n} -\log_2 \mathbf{M}(z_t \mid z_{<t}) \;\leq\; -\log_2 \nu(z) + K(\nu).$$

Proof. If $\nu(z) = 0$ the bound is trivial. Otherwise, for every index i of ν the mixture contains the term $2^{-K(i)}\nu(z)$, so $\mathbf{M}(z) \geq 2^{-K(\nu)}\nu(z)$ (Li and Vitányi, 2019, Ch. 4). The conditional log-losses telescope to $-\log_2 \mathbf{M}(z) + \log_2 \mathbf{M}(\varepsilon) \leq -\log_2 \mathbf{M}(z)$. □

The bound holds for every sequence, however it was produced, and it assumes nothing about the source. Its content is comparative: on any data, the mixture loses at most $K(\nu)$ bits more than any given model ν. The constant depends on the model and on the reference machine.

Proposition 7.2 (No computable predictor dominates)**.** *Let ρ be a predictor with computable conditionals satisfying $\sum_a \rho(a \mid z) \leq 1$. Write $K(\rho)$ for the least prefix complexity of a program computing ρ's conditionals. There is a computable sequence ω on which ρ's cumulative log-loss after n symbols is at least $n \log_2 |\Sigma| - O(1)$. Meanwhile the deterministic model of ω has log-loss 0 and complexity at most $K(\rho) + O(1)$. Hence no computable predictor satisfies a bound of the form of theorem 7.1.*

Proof. At each step, approximate $\rho(a \mid \omega_{<t})$ for every $a \in \Sigma$ to within $2^{-t}/|\Sigma|$ and let ω_t be the symbol with the least approximation, breaking ties in a fixed order. The least conditional is at most $1/|\Sigma|$, so $\rho(\omega_t \mid \omega_{<t}) \leq (1 + 2^{-t+1})/|\Sigma|$. Summing $-\log_2$ of these gives the bound, since $\sum_t \log_2(1 + 2^{-t+1}) < \infty$. The construction is uniform in a program for ρ. □

By premise P2, then, the benchmark is not a usable method: every usable predictor is computable, and for each there is a computable sequence on which it does as badly as possible. What theorem 7.1 supplies is a sense in which weighting models by their compressed length is an optimal bias up to $K(\nu)$: relative to every lower semicomputable model, on every sequence, it loses at most a constant. It is an ideal; usable approximations aim at it, and each is defeated on some computable sequence. Martins draws a related moral for physics from normative inductive reasoning: theories should be weighed, not chosen and discarded (Martins, 2017).

What compression offers. The benchmark nevertheless shows what compression is good for. Every model, a lookup table included, has outputs at settings not yet observed; what distinguishes a short model is its weight. The universal mixture gives a model of length $K(\nu)$ weight at least $2^{-K(\nu)}$, so by theorem 7.1 predicting as that model does costs at most $K(\nu)$ bits, while a lookup table pays about the length of the record. Compressing what is known therefore commits a theory cheaply to predictions about what is not yet known, and when those predictions are tested, the bias that chose the theory is put at risk.

- Maxwell's equations unified the known laws of electricity and magnetism. They then predicted electromagnetic waves traveling at the speed of light, which Hertz produced.
- Dirac's equation reconciled quantum mechanics with special relativity for the electron. It predicted a particle of the same mass and opposite charge, and the positron was found four years later.

- Compression can also fail. Le Verrier explained the anomalous precession of Mercury, within Newtonian mechanics, by an unseen inner planet, Vulcan. The planet does not exist, and the anomaly was explained by general relativity.

This is the value of unification, and it is why the search for compact theories of everything is worth pursuing: a more compact theory of more phenomena makes cheaper commitments beyond the record, and so it can be tested more severely.

Predictive success does not certify finality. Every success is a finite record, and by corollary 3.2 no finite record certifies that a model is final. Newtonian mechanics was confirmed by predictions for two centuries, including the discovery of Neptune, and it was not final. A theory of everything confirmed by every test anyone performs would be in the same position.

8 Objections and replies

We answer the ten objections that bear most directly on the main claim.

1. "This is just Hume, or Popper." Our first result, corollary 3.2, is a precise version of the Humean and Popperian point recalled in section 1, and we claim no novelty for it. The paper adds three things the Humean and Popperian traditions do not supply. First, a usable method need not *verify* a final theory to succeed: it could simply settle on one and stay there. Theorem 4.1 and remark 4.2 show that no usable method is guaranteed, even with any fixed probability, to do that on every computable world compatible with what has been seen. Any computable implementation of Popper's falsificationism is such a method, and it inherits the limit. Second, some underdetermination is permanent (objection 3); Popper's asymmetry between refutation and verification says nothing about rivals that can never be refuted. Third, the usual escape from underdetermination, preferring the simplest survivor, depends on the choice of universal machine (theorem 5.3). The Humean point is the first step of the argument, not its conclusion.

2. "A method need only work in our world, not in all worlds." Every method succeeds on some class of worlds and fails on others; we can grant that a method is reliable in ours. The question is how anyone could *know* that our world lies in its class. By the prefix form of theorem 4.1, every finite record—the whole of what anyone has observed, at any time—is consistent with computable worlds in which a usable deterministic method never settles (for a randomized method, worlds in which its chance of settling is as small as one likes). And for any method whose success class meets every record, such as one that identifies the eventually-zero worlds, the same record is also consistent with worlds in which it succeeds (lemma 4.6). The judgment that ours is one of the good worlds is therefore not drawn from the records. It is the method's bias, restated as a belief about the world. Belot argues that Bayesian convergence theorems make a similar confidence rationally mandatory (Belot, 2013).

3. "Underdetermination is only transient." Each rival in theorem 3.1 disagrees with the others somewhere, so each is refuted once the record reaches that setting. Sklar calls this "transient" underdetermination (Sklar, 1975, p. 380), and the objection is that it is only a temporary state of the evidence; Laudan and Leplin add that empirical equivalence is relative to a time and to auxiliary assumptions, and may later be broken (Laudan and Leplin, 1991). We grant both points rival by rival. Two things remain. First, at every stage infinitely many

unrefuted rivals still disagree about what comes next, and by the prefix form of theorem 4.1 no usable method can tell when the rivals that matter are gone: the transient phase never announces its end. Stanford argues that recurring transient underdetermination, in the form of alternatives no one has thought of, is the real threat to realism (Stanford, 2006); the rivals of theorem 3.1 show that unrefuted alternatives always exist, though they are not Stanford's unconceived theories. Second, some underdetermination is not transient at all. Models that agree on every possible observation are never separated, and chance hypotheses that agree on which events have probability zero are never told apart with certainty by any method (theorem 6.1).

4. "The rivals are cheap." Kukla gave algorithms that turn any theory into an empirically equivalent rival (section 3) (Kukla, 1993). Leplin and Laudan reply that such constructions are not genuine scientific rivals (Leplin and Laudan, 1993); Kukla later defended them (Kukla, 1996); and Stanford that rivals produced by a general recipe amount to Cartesian skepticism (Stanford, 2001). Three replies. First, the criteria that exclude cheap rivals—independence, naturalness, explanatory depth—are exactly the kind of preference lemma 4.6 identifies: real, often useful, and not supplied by the records. Second, the rivals need not be cheap. Proposition 5.7 gives rivals within $K(x) + O(1)$ bits of the shortest model that predict a different outcome at any unobserved setting, so the excluding criterion cannot be length. Third, physics supplies rivals no one regards as artificial: Newtonian, Lagrangian and Hamiltonian mechanics; matrix and wave mechanics; dual descriptions of the same physics; and the interpretations of quantum mechanics (sections 5 and 6). Nor is the argument skeptical: it posits no deceiver and holds for any record, however it was produced.

5. "Simplicity, or beauty, tracks the truth." The hope for a final theory is usually a hope that the true theory is simple and beautiful (Weinberg, 1992). Five points.

- *Description length.* The most precise notion of simplicity depends on the choice of universal machine (theorem 5.3): on any finite record, any consistent model can be made the shortest, at a cost of about $K(e)$ bits, so the record leaves open the order of candidates within that margin. Description length is called objective because it is invariant up to an additive constant. We agree, both asymptotically and for comparisons separated by more than that constant; but near-tied candidates that disagree about the unobserved always exist, and a change of machine that moves no description length by more than about the gap makes any of them the shortest (proposition 5.7 and theorem 5.5).

- *Ideal MDL.* Vitányi and Li (2000) show that ideal MDL is Bayes-optimal when the data are random for the true hypothesis. We accept this. It is a conditional guarantee whose condition, typicality, no finite record can confirm, since randomness deficiency is uncomputable; and it yields convergence in the limit, never certification.

- *Hard to vary.* Deutsch holds that good explanations are those whose details cannot be varied without spoiling their fit, and that choosing them is rational, not a matter of taste (Deutsch, 2011, ch. 1). Taken informally, this is one more constraint in the sense of lemma 4.6; formalized as a count of free parameters or a description length, it inherits machine-relativity. But Deutsch also denies that any explanation is final: "there can be no such thing as an ultimate explanation" (Deutsch, 2011, ch. 3). His view does not oppose the main claim. He opposes us only on whether the selection is rational, and nothing in our argument requires that it be irrational, only that the records do not make it.

- *Ockham efficiency.* Kelly and Genin prove that an Ockham method, which prefers the simplest answer to a fixed question, converges to the truth with worst-case optimal retractions (Kelly, 2007), or optimal reversals and cycles of opinion (Genin and Kelly, 2019). This justification is not machine-relative, and we accept it. It is relative to a chosen question and its partition into answers, and it guarantees efficient convergence in the limit, never certification: an Ockham method never knows that it has stopped retracting. It is consistent with our claim.
- *Beauty and naturalness.* Our point is independent of Hossenfelder's diagnosis (section 1): such criteria are constraints the records do not supply, and no record can confirm that the world satisfies them.

6. "The success of physics shows convergence on the truth." The no-miracles argument holds that the predictive success of mature theories would be a miracle unless they were approximately true (Putnam, 1979). We accept the success and explain it differently. A compact model of what is known commits itself about what is not, and when those commitments survive test the model has earned its place (section 7). The ideal predictor of theorem 7.1 enters only as a benchmark: it shows that weighting models by compressed length is a bias that loses little against any computable model, but no scientist uses it, and it does not explain why particular theories succeed. The success of physics is evidence that its theories are good compressions; it cannot certify that any of them is the last (corollary 3.2 and section 7).

7. "Algorithmic probability is itself the law." Müller's algorithmic idealism (Müller, 2026) shares our starting point and our tools: it begins from the observer's record rather than from a world, and it is formalized within algorithmic information theory (Müller, 2020). It also anticipates theorem 5.3. In *Law without law* (Müller, 2020) Müller holds, citing his own result that the most natural attempt to privilege a universal machine fails (Müller, 2010), that the choice of algorithmic prior is arbitrary, and he restricts the theory's predictions to those that are identical for every choice. From these postulates he derives that observers will, with high probability, come to see an external world governed by simple computable laws. The step we decline is his postulate that algorithmic probability *gives the chances* of what an observer sees next, which makes it the fundamental law. Read as a candidate final theory, this postulate is subject to three limits. No record certifies it: every finite record has positive probability under infinitely many computable measures that disagree about what comes next (theorem 3.1(c)), so by part (b) of corollary 3.2 any certification of the universal mixture at a record is wrong in some computable world that produces that record. Adopting it is a choice of bias (lemma 4.6), and Müller's own restriction to machine-invariant predictions concedes that single-step chances are not fixed by the theory; its machine-invariant predictions are asymptotic. And the mixture is uncomputable, so no usable agent can apply the law it would be (proposition 7.2). Algorithmic idealism is therefore not a counterexample to our claim but an instance of it: a constraint, explicitly chosen, that no observation could confirm as final. The two views agree on what records can and cannot settle, and differ only on whether to affirm a law beyond them. Müller affirms one; we remain agnostic.

8. "The second premise begs the question." Refuting premise P2 requires a *usable* hypercomputer: a system whose non-computable behavior can be specified, relied on and put to work, not one whose user cannot rule out spurious signals or device failure (Earman and

Norton, 1993). For deterministic methods C2 survives any hypercomputer with the power of a Turing oracle, provided theories can use the same power, because the diagonal argument reruns at every level of computing power (theorem 4.5 and premise P2*). This also answers Faizal and colleagues (section 1): non-algorithmic understanding would help only if it were unavailable to the theories it was used to choose among.

9. "Knowledge needs only reliability, not proof." On fallibilist and externalist accounts, a belief can be knowledge because it was reliably produced or is safe, without the record ruling out every rival. The externalist may say that a reliable method that has settled on the final theory, in a world where it is final, yields knowledge in their sense. We need not deny this: a believer in that world may hold the truth. What no one can do is establish, from the records, that they are in that world (C1, C2, lemma 4.6), and that is what "known to be final" means here (definition 2.6).

10. "The argument undermines itself." Premises P1 and P2 are universal empirical claims, so, as for any universal claim, no record certifies them either. We do not claim to know them. We assert them as premises and state what would refute each (section 2 and premise P2*; objection 8), so every conclusion is conditional on them. This is consistent: the paper claims no certification of a kind it shows to be impossible.

9 Conclusion

We asked whether a theory of everything could ever be known to be final, and argued that it could not.

(C1) No method of any kind can certify a theory as final: if a method certifies a model, deterministic or stochastic, at a record, a computable world that produces that record makes the certification wrong (corollary 3.2).

(C2) No usable method is guaranteed, even with any fixed probability, to settle on a final theory. Whatever has been observed so far, there is a possible world that fits all of it in which a deterministic usable method never settles on the right theory (theorem 4.1), and for a randomized one and each $p > 0$, there is a world in which it settles with probability less than p (remark 4.2). This part rests on the second premise that usable methods are computable (premise P2), and for deterministic methods only on its weaker form (theorem 4.5).

(C3) Even an ideal agent cannot separate formulations with identical predictions, and cannot tell apart with certainty chance hypotheses that agree on which events have probability zero (theorem 6.1).

As a corollary (C′), simplicity narrows the field but cannot decide among the survivors that matter. The data fix which models fit; which of them is shortest depends on the reference machine. Rivals that fit equally well, are nearly as short and disagree about the unobserved always survive, and a change of machine that moves no description length by more than about a rival's gap makes it the unique shortest (theorem 5.5 and corollary 5.8). The universal mixture still predicts within $K(\nu)$ bits of every model ν (theorem 7.1). Every choice among them is a constraint we impose (lemma 4.6).

The position is agnostic in Huxley's sense. It does not deny that the world has a true description, or that physicists might one day write one down; it denies that anyone could know, from the records, that they had. Nor does it counsel giving up on unification. Compact theories remain worth seeking, because they compress what is known and commit to new, testable predictions (section 7). What the argument removes is the goal of a final theory known to be final, and with it the hope that simplicity, beauty or unification could certify one.

Put plainly: we can keep finding better theories of everything, but we can never know, from the records, that we have found the last one.

Statement on the use of AI

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